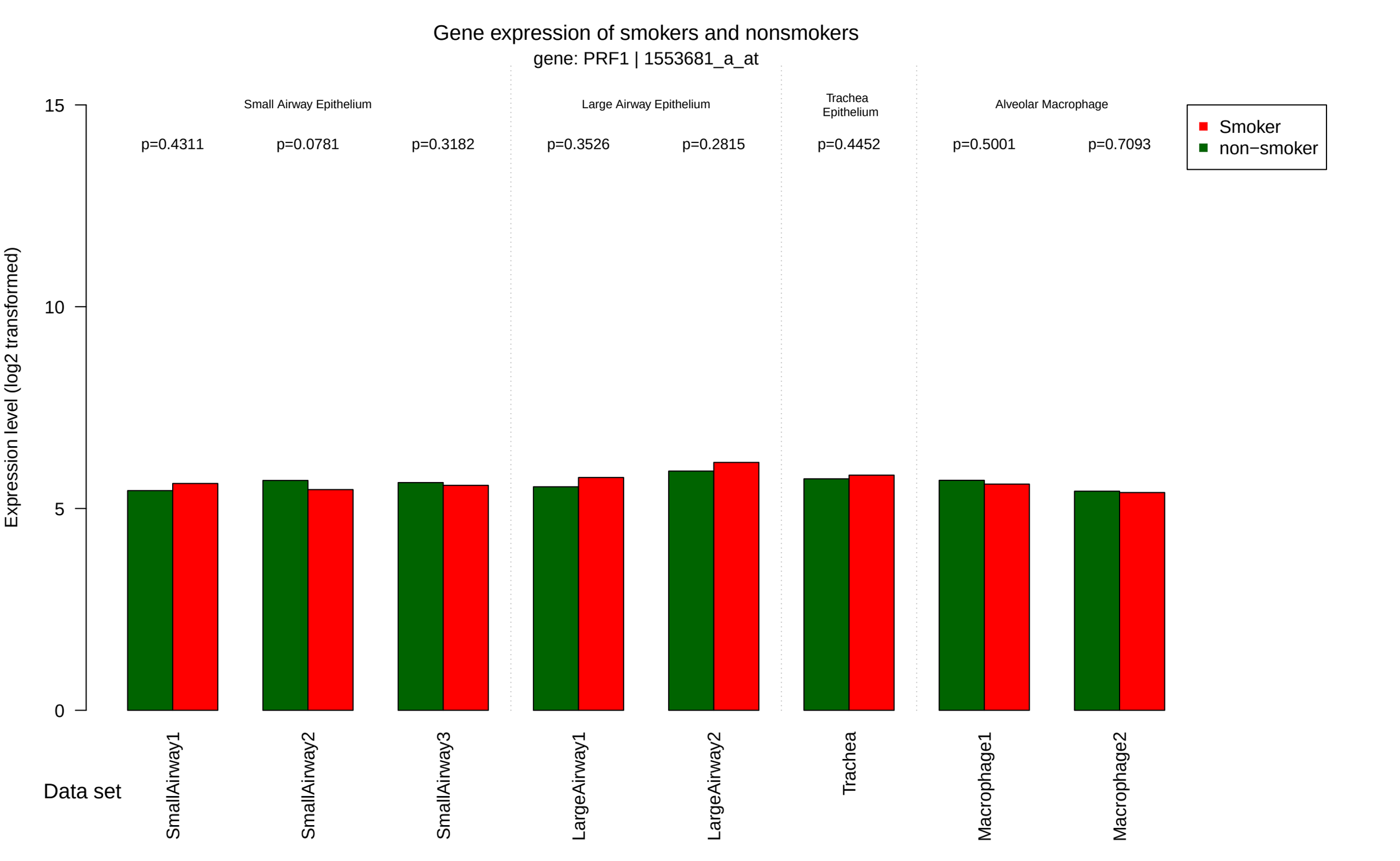
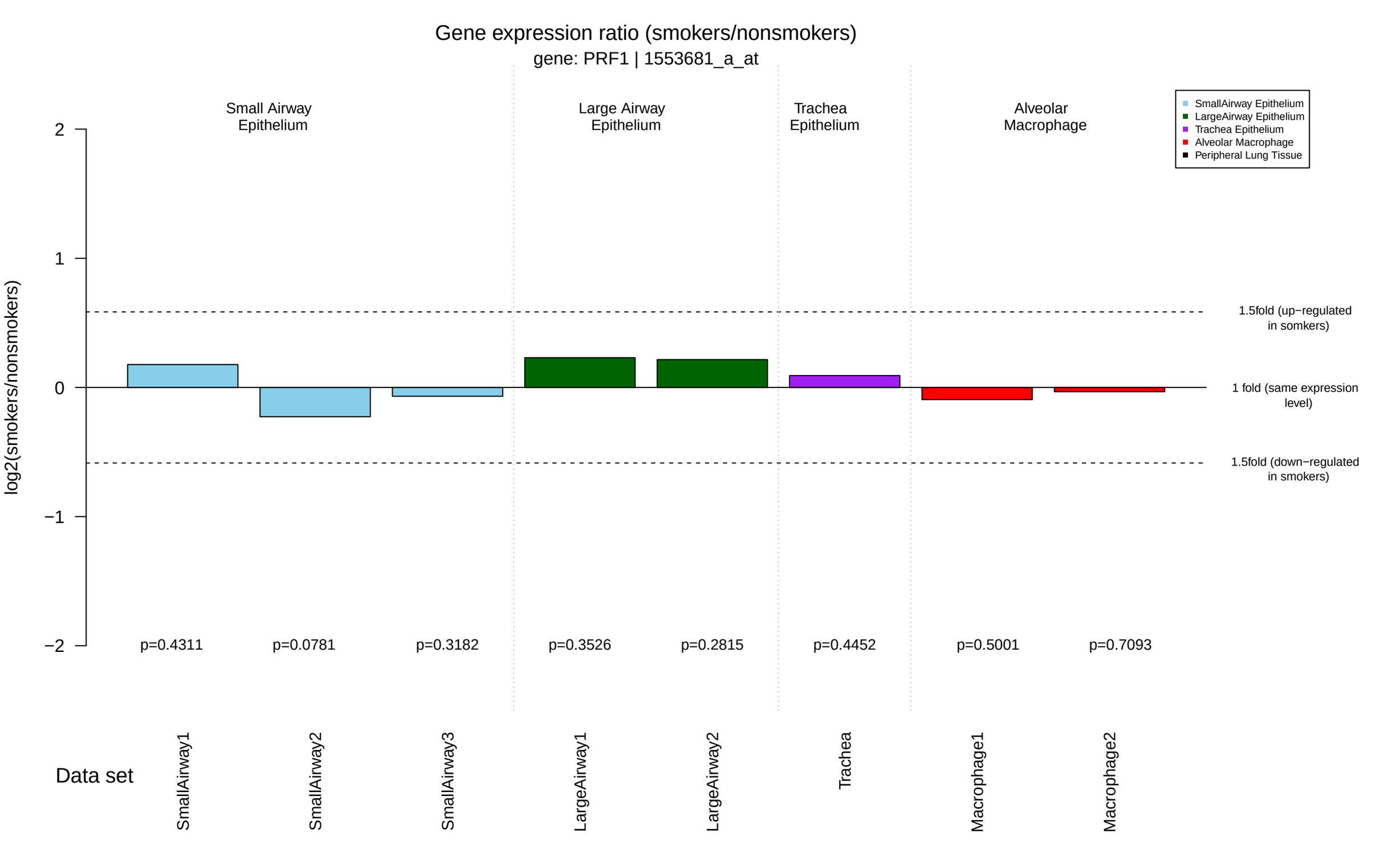
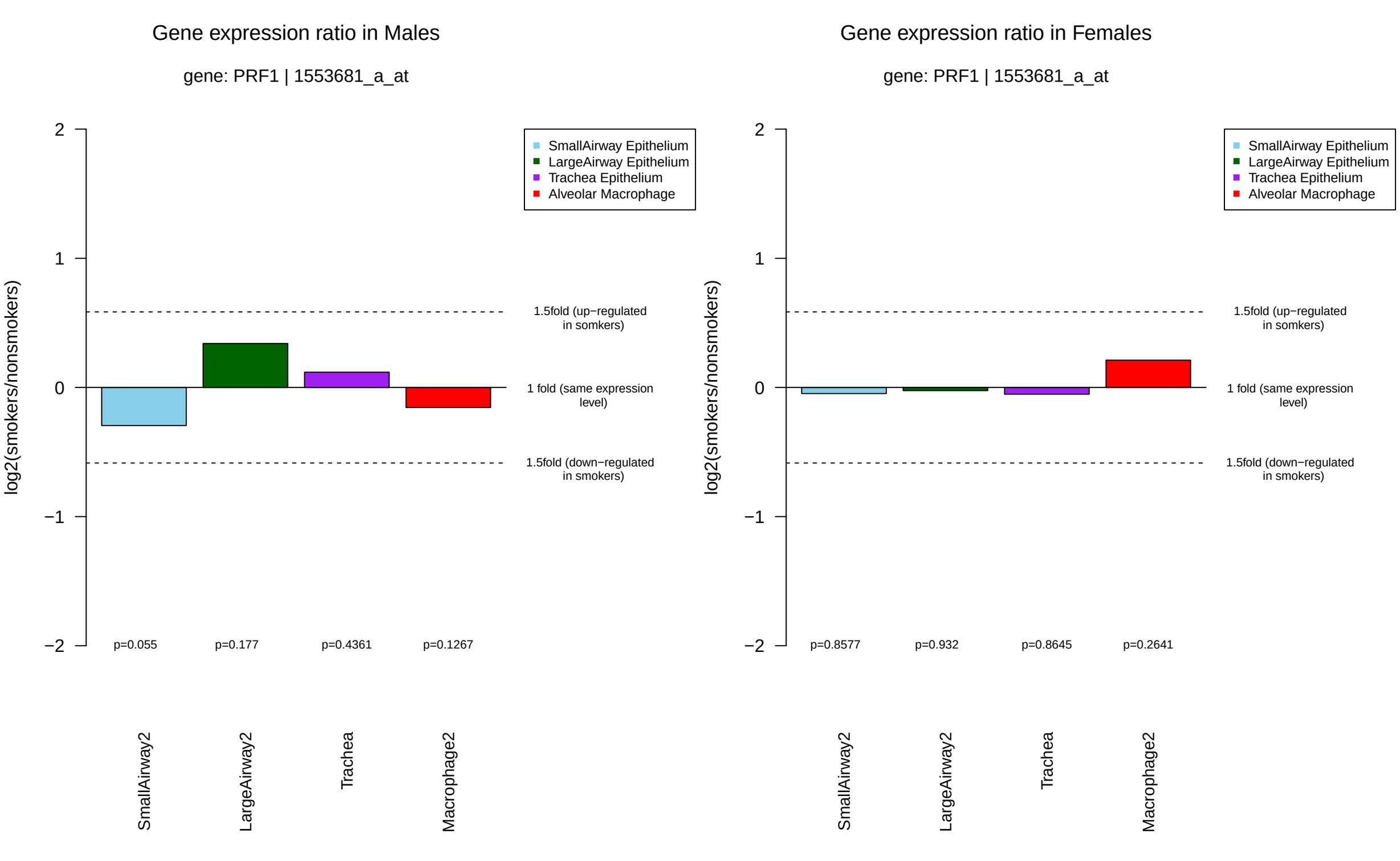
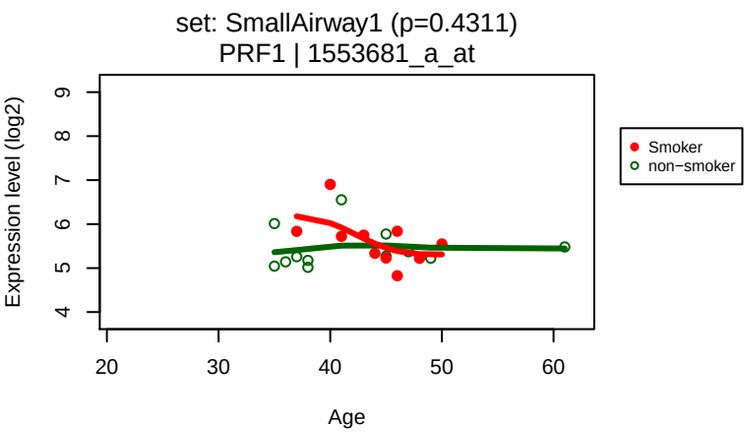
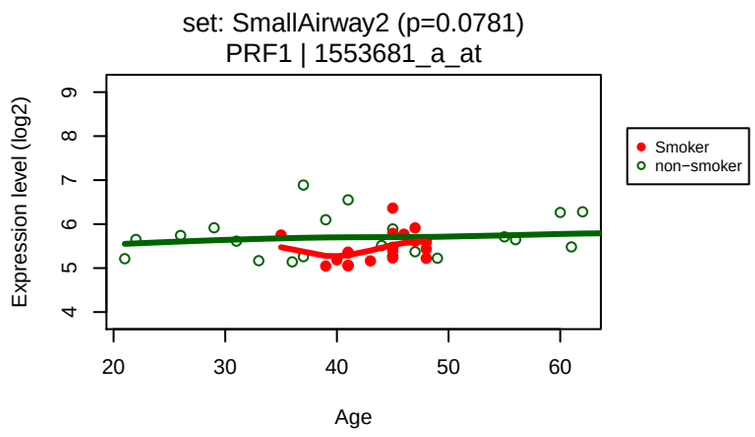
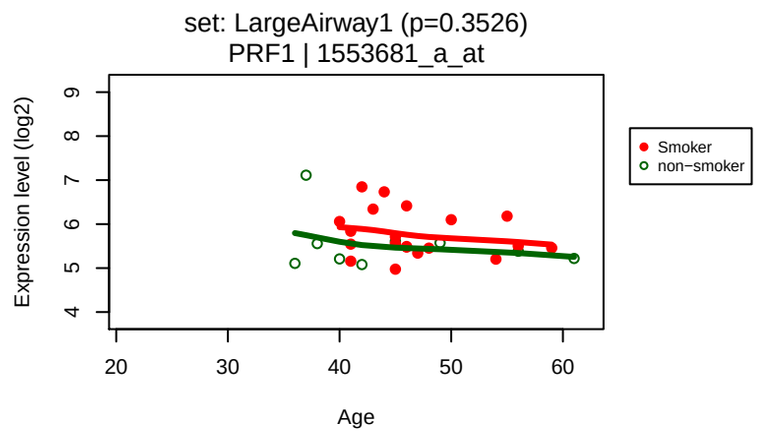

Gene expression of smokers and nonsmokers
gene: PRF1 | 1553681_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4311
p=0.0781
p=0.3182
p=0.3526
p=0.2815
p=0.4452
p=0.5001
p=0.7093
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

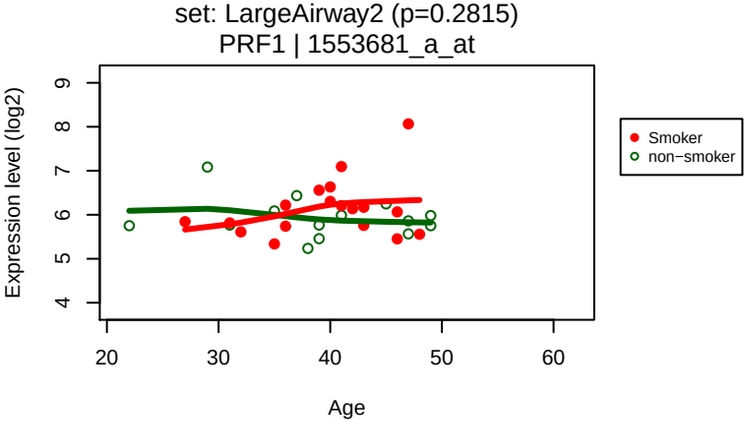

Gene expression ratio (smokers/nonsmokers)
gene: PRF1 | 1553681_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4311
p=0.0781
p=0.3182
p=0.3526
p=0.2815
p=0.4452
p=0.5001
p=0.7093
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

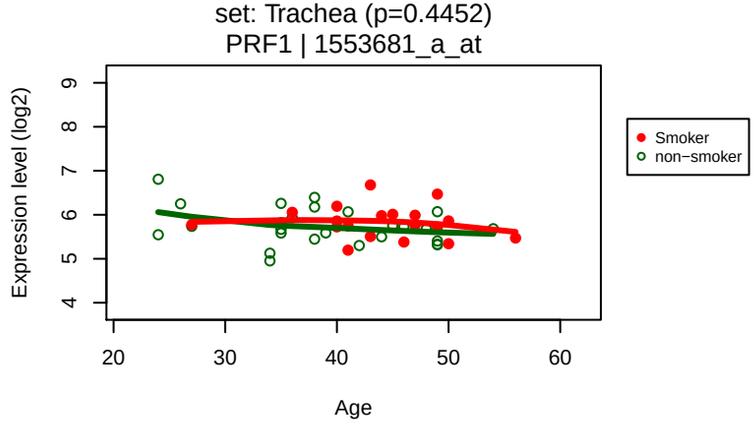

Gene expression ratio in Males
gene: PRF1 | 1553681_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.055
p=0.177
p=0.4361
p=0.1267
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRF1 | 1553681_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8577
p=0.932
p=0.8645
p=0.2641
SmallAirway2
LargeAirway2
Trachea
Macrophage2

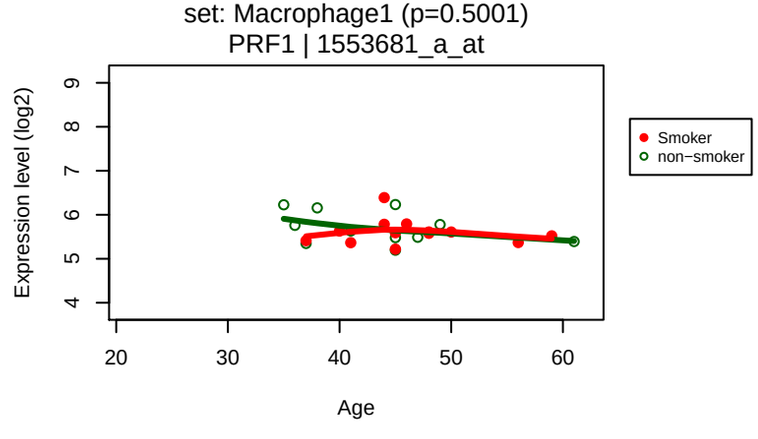

set: SmallAirway1 (p=0.4311)
PRF1 | 1553681_a_at
Expression level (log2)
Age
Smoker
non−smoker

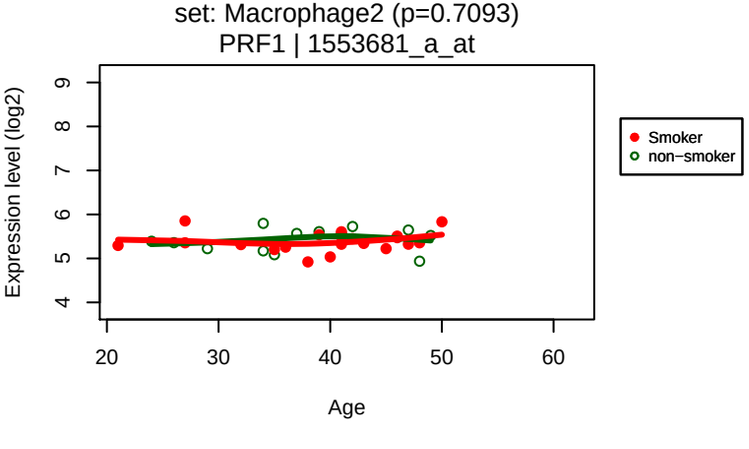

set: SmallAirway2 (p=0.0781)
PRF1 | 1553681_a_at
Expression level (log2)
Age
Smoker
non−smoker

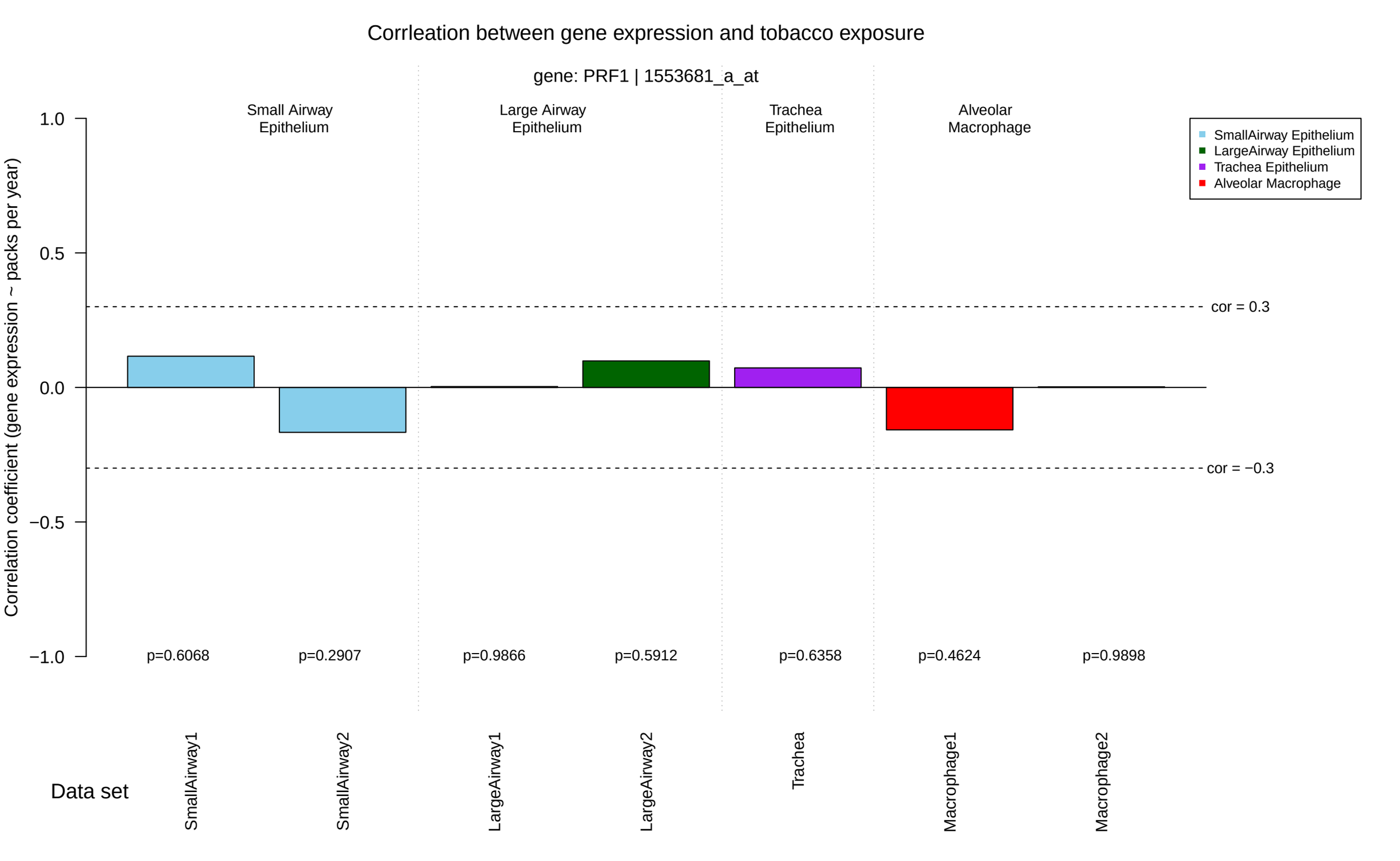

set: LargeAirway1 (p=0.3526)
PRF1 | 1553681_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2815)
PRF1 | 1553681_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4452)
PRF1 | 1553681_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5001)
PRF1 | 1553681_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7093)
PRF1 | 1553681_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRF1 | 1553681_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6068
p=0.2907
p=0.9866
p=0.5912
p=0.6358
p=0.4624
p=0.9898
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2